

# Josh Rehders

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## PROFESSIONAL EXPERIENCE

- **Microsoft** | Software Engineer 2024 – Present
  - Shipped Copilot Page/Notebook experiences in Microsoft [365 Copilot](#) (100M MAU), [Teams](#) (300M), and [Loop](#) (25M) iOS apps.
  - Built [Copilot Notebook Sharing](#), letting users unify references and instructions across collaborators' AI generations.
  - Raised [Copilot Pages](#) return rate by surfacing recent pages from Chats and Notebooks in the M365 Copilot app.
  - Drove monitoring/observability for Copilot Pages/Notebooks in the three apps, catching regressions before user impact.
  - Unblocked Loop adoption by Fortune 500 customers by adding support for Microsoft [Purview Information Protection](#).
  - Built the Push Notification settings functionality in [Loop](#) iOS, and onboarded new notification signals into the system.

## EDUCATION

- **Vanderbilt University** | Nashville, Tennessee & Madrid, Spain Class of 2024
  - **Bachelor of Science in Computer Science**, Minors in Engineering Management, Data Science.

## PROJECTS

- **flight-tracker** | Personal flight price tracker + [Mother's Ruin Mother's Day Challenge](#) route solver
  - Built FastAPI + React + SQLite app with scheduled price polling, Firebase push on price drops, and a multi-city itinerary enumerator that solved the [Mother's Ruin Mother's Day Challenge](#) route puzzle.
  - Deployed via GitHub Actions (Docker, Caddy, GHCR), built with IP rate limiting and CORS lockdown for safe public usage.
  - Contributed two new filtering features to [fli](#), the open-source library used for advanced filtering and flight-data analysis.
- **mrmdc-backend / mrmdc-validator** | Backend for Apple Wallet [Mother's Day Challenge](#) cards + card validator for staff
  - Built the backend in Node 20 / Fastify / Postgres with Apple PassKit + Google Wallet passes and Resend transactional email.
  - Shipped the SwiftUI validator app — scans a customer's pass and shows valid/invalid status in under 600 ms.
- **mrmdc-passport** | Mobile companion app for the [Mother's Ruin Mother's Day Challenge](#)
  - Built SwiftUI app with location-verified check-ins (CoreLocation + on-device Haversine geofence) and a network-resilient background queue that drains pending stamps without blocking the UI.
  - Added a split-flap countdown graphic for the user's next flight, shown alongside the 5-venue grid and progress tracker.
- **photo-admin / photos-delivery** | Booking + delivery tooling for freelance creative services
  - Photo-admin: Next.js + Postgres to aggregate upcoming NYC shows I may be interested in based on my Spotify + Stats.fm data, and gives the ability to send templated email outreach to artist contacts with one click.
  - Photos-delivery: Next.js + Cloudflare Workers + R2/D1, with multipart upload of large videos for performance. Public client galleries with optional passwords, streaming zip downloads, and rate-card driven invoice generator based on shoot details.
- **photo** | Creative media portfolio at [rehders.photos](#)
  - Built Astro + Tailwind static site that delivers galleries from Cloudinary CDN, deployed to Cloudflare Pages.

## OTHER INVOLVEMENTS

- **Rehders Photos** | Freelance Multimedia Creative Specialist 2020 – Present
  - Created photo and video content for artists, venues, and publications nationwide including concerts, festivals, and sports.
  - Managed travel logistics for shoots, including transportation and accommodation in 10+ cities across the country.
  - Covered ARC Music Festival, Lollapalooza, College World Series, March Madness, CFP National Championship, and more.
- **The Vanderbilt Hustler** | Photography Director 2020 – 2024
  - Shot 300+ [assignments](#) for [student publication](#); taught Lightroom workflows to staff for consistent content quality.
- **Swipes for a Cause** | Co-Founder and Public Relations Manager 2020 – 2024
  - Co-founded [excess meal donation program](#) during Fall 2020, temporarily shutdown by administration shortly after launch.
  - Restructured into a [Campus Dining partnership](#) that collected 5,000 meals and \$2,000 of cash donations per semester.

## SKILLS, INTERESTS AND AWARDS

- **Languages:** Swift, TypeScript, Python, Objective-C/C++, C++, Java, C#.
- **Frameworks:** SwiftUI, UIKit, React, Next.js, Fastify, FastAPI.
- **Infrastructure:** Docker, GitHub Actions, GCP, Cloudflare Workers/R2/D1, Postgres, SQLite, Firebase Cloud Messaging.
- **Interests:** Aviation (300+ flights), concert photography (40+ shows), international travel (24 countries), vinyl records, live music, skiing, college baseball, craft coffee, home cooking.
- **Awards:** Eagle Scout, National Hispanic Scholar, Dean's List, CMA Pinnacle (3x), Associated Collegiate Press Story of the Year.